

The Economics of Artificial Intelligence

In 1589, Queen Elizabeth I famously refused a patent grant for a knitting machine because she feared her people would become "beggars." The industrial era eventually displaced routine manual tasks with capital. Artificial Intelligence (AI) is the modern manifestation of the "Elizabethan Anxiety," as it enables the replacement of non-routine cognitive tasks with capital.

What makes AI unusual is not its power, but how quickly it is spreading across industries (Damodaran, 2026). This poses the risk of a "Great Decoupling," in which the inability to adapt quickly widens the gap between those who control AI and those displaced by it. Skill-Biased Technological Change (SBTC) illustrates that AI is not a neutral force. It instead accelerates labor-market polarization, endangering the middle class while benefiting those at the forefront of technology.

AI complements high-skilled workers engaged in non-routine, cognitively demanding tasks like specialized engineering, strategic management, and medicine. According to the Task-Based Model, a job is a "bundle" of tasks, and AI frees professionals to focus on "high-value" tasks such as complex problem-solving and idiosyncratic judgment by automating "routine cognitive" tasks, such as data cleaning, literature reviews, and basic drafting (Acemoglu & Autor, 2011). This efficiency need not necessarily result in job displacement, as explained by Jevons' paradox (Alcott, 2005). AI thus raises the marginal productivity of skilled labor, resulting in capital-skill complementarity.

People, however, often overlook the crucial detail of "value capture." Even if AI makes a programmer twice as productive, it doesn't guarantee a proportionate increase in their pay. If AI becomes a commodity and the elasticity of substitution is high, "producer surplus" could be transferred as "consumer surplus." Businesses that use AI are more productive, but only people who can offer judgment beyond AI's capabilities can capture that value (Hampole et al., 2025). Therefore, to keep their "pricing power," high-skilled workers need to master the Centaur model of human-AI collaboration. As the machine handles the routine, the human provides the 'last mile' of judgment, requiring that unique, human-centered knowledge that AI models can't commoditize.

The outlook for middle-skilled workers, like administrative managers, accountants, and paralegals, is much worse. Routine-Biased Technological Change (RBTC) affects these jobs because they are rule-based, and AI acts as a direct substitute rather than a productivity-enhancing complement.

The "Citrini Scenario" is a tail risk in which AI takes jobs away from people faster than the economy can create new ones (Citrini Research, 2026). IMF's data from 2024 shows that this demographic faces the highest risk of displacement due to AI. The labor market becomes "occupationally polarized" when we eliminate these "bridge" jobs, which offer a pathway for

upward social mobility. If the rate of displacement exceeds the rate of human retraining, the former middle class will have less money to spend, resulting in a macroeconomic crisis.

Contrary to popular belief, low-skill manual jobs like maintenance, hospitality, and elder care are "biologically" shielded from AI but economically vulnerable. These jobs require "non-routine manual" skills that are difficult for AI and robotics to imitate at low cost. However, they are vulnerable to the cascading effect from middle-skilled workers if rapidly displaced by AI. The shift of labor supply towards the left will create a "U-shaped" labor market, directly worsening wage inequality.

The Human Capital Theory explains the widening economic divide between people with and without specialized degrees (Becker, 1993). When the elasticity of substitution between AI and middle-skill labor is high, the "labor share" of national income decreases while the "capital share" increases. In essence, the people who own the algorithms or have the exceptional skill to manage them profit from AI-driven productivity, while those who can't fall behind. Given the speed of development and the scale of potential displacement, we need policy interventions both for the short and the long term.

As a short-term measure, we need to address the asymmetries in the U.S. tax code. Acemoglu and Restrepo (2020) state that the current fiscal framework is fundamentally unfair, taxing workers at high payroll rates while effectively subsidizing automation through generous rules on capital depreciation. The Mercatus Center (2025) supports my idea of Human Capital Tax Parity, which would allow businesses to write off the costs of training their workers right away. We should also strengthen the "experience rating" in unemployment insurance so that companies take into account the social costs of rapid displacement. Ratner (2013) showed that making taxes more sensitive to companies that hire and fire workers makes them consider the productivity of a "bot" relative to the cost to society of an unemployed worker. Lastly, a neutrality-based payroll credit would fix the "human disadvantage" by making the tax burden the same for a \$20/hour person and a computer. Conceptually, this functions as a Pigouvian tax, a levy designed to correct a market failure. By taxing the negative externality of sudden labor displacement, the government can fund the retraining needed to reabsorb those workers into the economy.

As a long-term measure and to narrow the "speed gap" between AI evolution and human adaptation, the United States needs to move beyond the entrenched four-year degree model, which is a one-size-fits-all solution. I advocate the creation of Federally-Funded Lifelong Learning Accounts (LLAs) in which American workers in sectors exposed to high AI-displacement risk would receive a portable, tax-advantaged account, seeded with government credits that could only be used at accredited AI-certification, vocational-retraining, or skill-upgrading programs. LLAs facilitate Schumpeter's Creative Destruction, by providing liquid capital required to transition displaced workers into AI operations

But LLAs face two major obstacles. The first is a quality-control risk, which, without rigorous oversight, could be exploited by low-value “certification mills.” Second is the fiscal cost. To prevent adding to the national deficit, they could be financed by a “Digital Productivity Tax” on businesses that experience benefits from automation. The detractors argue that such a tax may hurt innovation, but a socially sustainable innovation is better than one that creates greater economic fracture.

Just like Elizabeth’s fear about the knitting machine was unfounded and gave birth to the textile industry, the doomsday scenario of AI-led displacement is not a certainty. It’s clear the risk is not the “machine” but our lack of policy support. Through proactive policy tools such as LLAs and tax parity, we can transform the “Elizabethan Anxiety” of our time into renewed investment in the most versatile skill of all: human judgment.

Bibliography

Acemoglu, D., & Autor, D. (2011). Skills, tasks and technologies: Implications for employment and earnings. *Handbook of Labor Economics*, 4, 1043-1171.

Acemoglu, D., Manera, A., & Restrepo, P. (2020). *Does the US tax code favor automation?* (Working Paper No. 27061). National Bureau of Economic Research.
<https://doi.org/10.3386/w27061>

Alcott, B. (2005). Jevons' paradox. *Ecological Economics*, 54(1), 9–21.
<https://doi.org/10.1016/j.ecolecon.2005.03.020>

Becker, G. S. (1993). *Human capital: A theoretical and empirical analysis, with special reference to education* (3rd ed.). University of Chicago Press.

Citrini Research. (2026, February 22). The 2028 global intelligence crisis: A thought exercise in financial history, from the future. Substack. <https://www.citriniresearch.com/p/2028gic>

Damodaran, A. (2026). AI Scenarios: From Doomsday Destruction to Do-Nothing Bots! [Musings on Markets].
<https://aswathdamodaran.blogspot.com/2026/03/ai-scenarios-from-economic-doomsday-to.html>

Hampole, M., Papanikolaou, D., Schmidt, L. D. W., & Seegmiller, B. (2025, February). Artificial intelligence and the labor market (Working Paper No. 33509). National Bureau of Economic Research. <https://www.nber.org/papers/w33509>

International Monetary Fund. (2024). Gen-AI: Artificial intelligence and the future of work. IMF Staff Discussion Note.
<https://www.imf.org/en/Publications/Staff-Discussion-Notes/Issues/2024/01/14/Gen-AI-Artificial-Intelligence-and-the-Future-of-Work-542316>

Mercatus Center at George Mason University. (2025). *A proactive response to AI displacement: The case for human capital tax parity.* [Policy Brief].

Ratner, D. (2013). *Unemployment insurance experience rating and labor market dynamics.* Federal Reserve Board: Finance and Economics Discussion Series.